

Application: SD for Generative Modeling

Learning Generative Models with Sinkhorn Divergences

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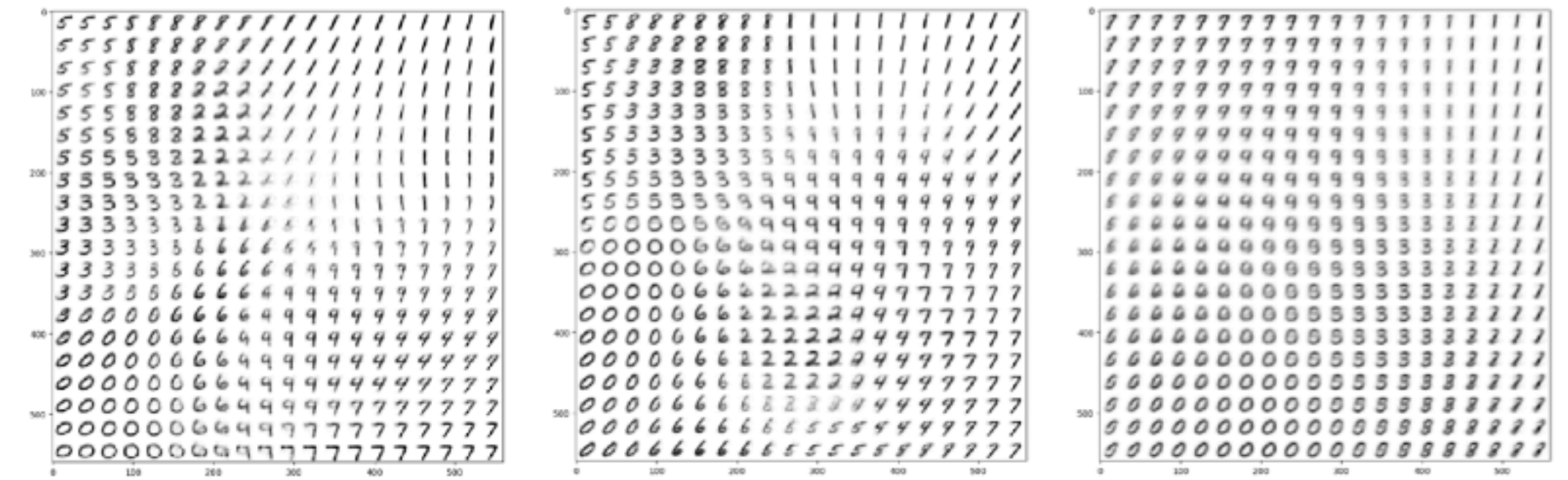
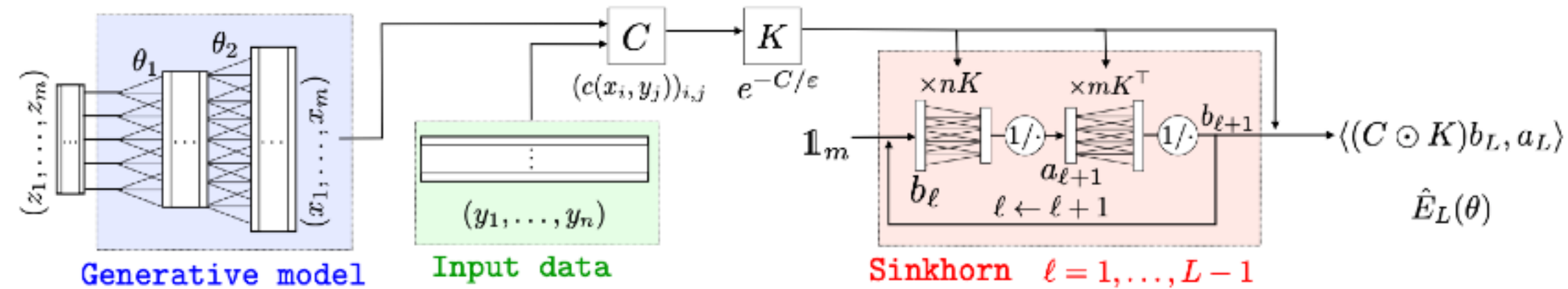
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Abstract

The ability to compare two degenerate probability distributions, that is two distributions supported on low-dimensional manifolds in much higher-dimensional spaces, is a crucial

1 Introduction

Several important statistical problems boil down to fitting densities, *i.e.* estimating the parameters of a chosen model that *fits* observed data in some meaningful way. While the standard approach is maximum likelihood estimation, this approach is often flawed in

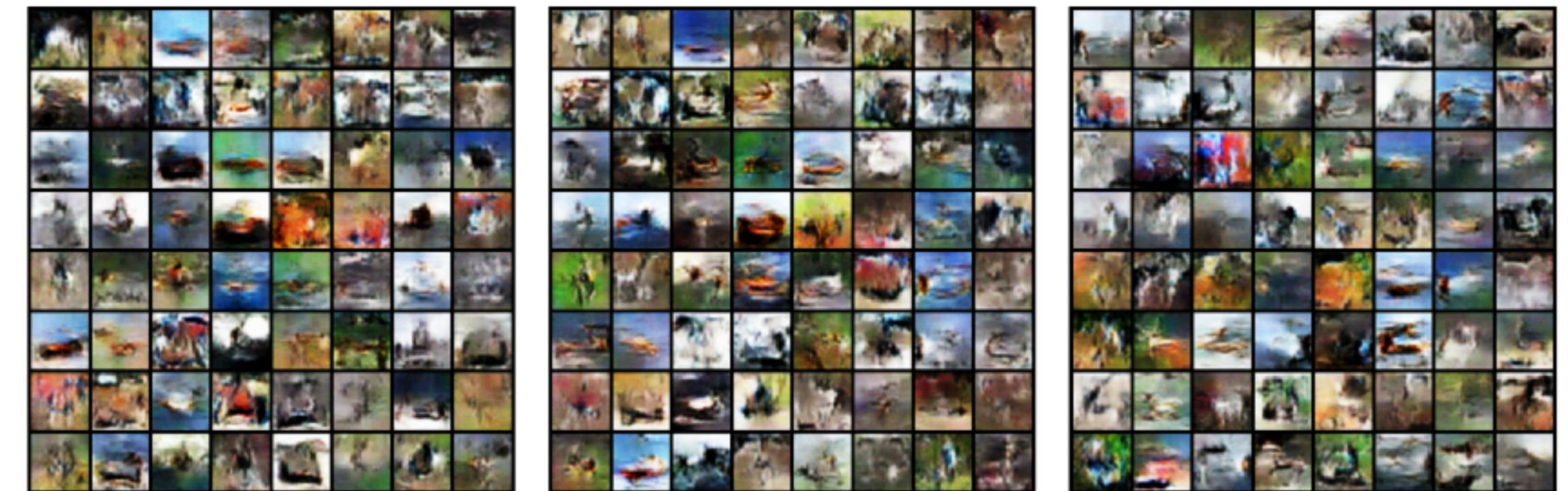


(a) $\epsilon = 1$
 $m = 200, L = 10$

(b) $\epsilon = 10^{-1}$
 $m = 200, L = 100$

(c) $\epsilon = 10^{-1}$
 $m = 10, L = 300$

Figure 3: Influence of the hyperparameters on the manifold of generated digits.



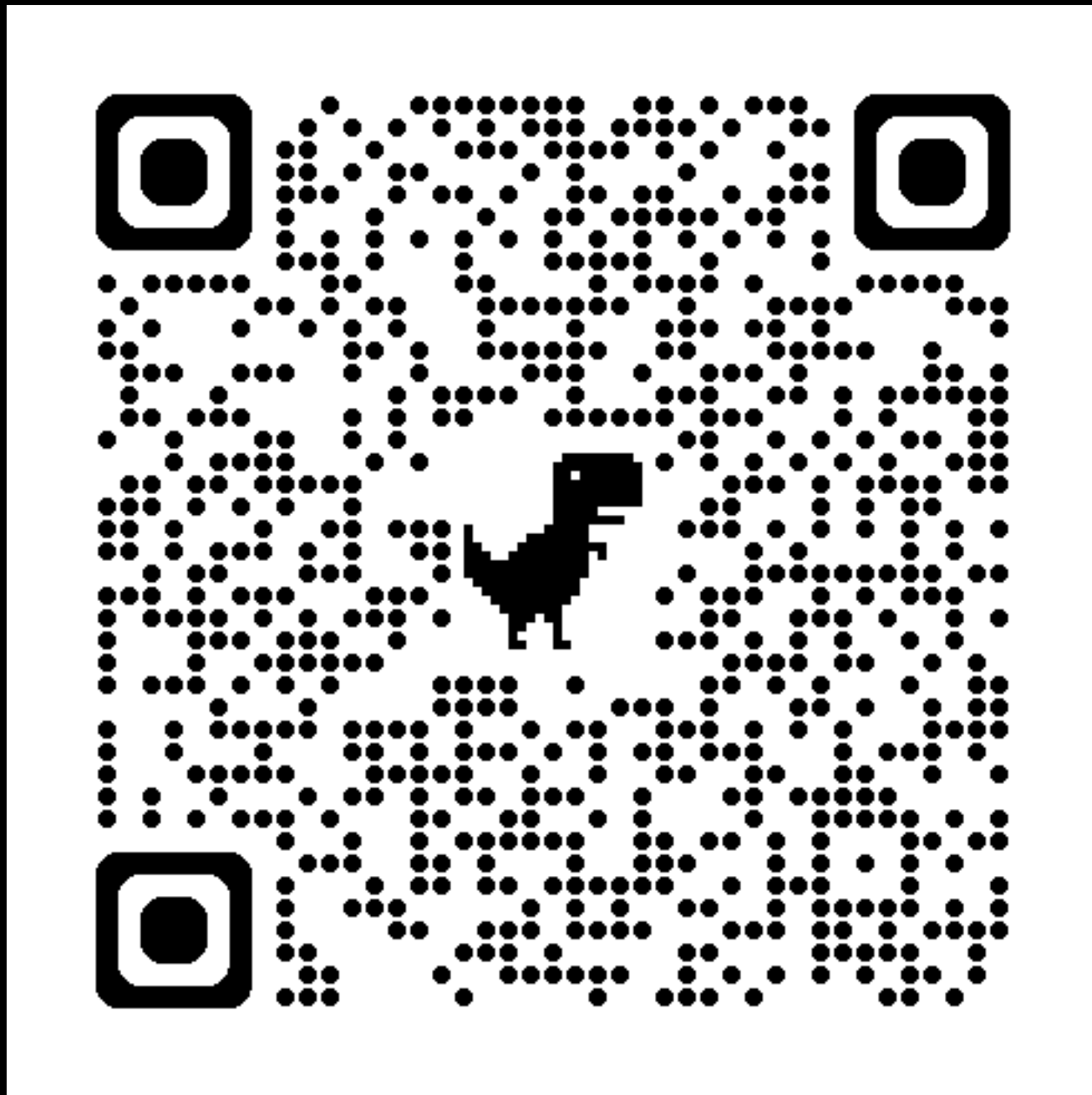
(a) MMD

(b) $\epsilon = 1000$

(c) $\epsilon = 10$

Interactive Demo: EOT Mapping/Loss

1. Compute couplings using exact and entropy-regularized OT (via POT)
2. Implement barycentric projection from a coupling
3. Implement sinkhorn divergence using POT's differentiable primitives



$$\tilde{T} : \mathbf{x}_i \mapsto \frac{1}{a_i} \sum_j \pi_{ij} \mathbf{y}_j$$

$$S_\varepsilon(\alpha, \beta) = \text{OT}_\varepsilon(\alpha, \beta) - \frac{1}{2} \text{OT}_\varepsilon(\alpha, \alpha) - \frac{1}{2} \text{OT}_\varepsilon(\beta, \beta)$$

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